

On-Board Charger (OBC) AC Grid & Thermal Test Report

11 kW Power Factor, Control Pilot PWM, Earth Leakage, and Thermal Cutoff

1. AC Grid Charging Power & Power Factor

- Test Case TC-OBC-031: Grid test at 400 V AC 3-phase delivered 11.05 kW with measured power factor of 0.992. Verdict: PASS.
- Test Case TC-OBC-032: Control pilot duty cycle sweep from 10.0% to 90.0% measured with maximum error of $\pm 0.22\%$ duty cycle. Verdict: PASS.

2. Wide-Range AC Input & V2L Bidirectional Discharge

- Test Case TC-OBC-033: Tested at 230 V AC / 50 Hz and 120 V AC / 60 Hz. Low-line voltage limit (85 V) and frequency margin (47 Hz) tested. Verdict: PASS.
- Test Case TC-OBC-034: V2L evaluated with 1.8 kW resistive heater: output 230.4 V, THD 1.8%. Inductive load testing (3.6 kW motor load) tested. Verdict: PASS.

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3. Touch Leakage Current & Thermal Shutdown Limit

- Test Case TC-OBC-035: Earth leakage current test: measured touch leakage current was 5.2 mA rms due to Y-capacitor filter sizing, e
- Test Case TC-OBC-036: Inlet temperature override test: controller throttled at 95.0°C and did not terminate charging until 108.0°C.